Study notes: Section 1: growth and decay of current in an L-R circuit (Murugeshan 12.1, 12.2)

Murugeshan, *Electricity and Magnetism* (10th ed.), Ch. 12 §12.1 and §12.2 (Module 4, syllabus item 15–16). Figures are drawn for these notes from the book formulas; the book figure number is in each caption. Problems are not included.

12.1 Growth of current in a circuit with resistance and inductance

Source: book-sources/m4-s12-1-growth.html, Murugeshan pp. 312–313 (printed). Syllabus item 15. Equations (1)–(4) of §12.1 are given in full.

Before you start

- Ohm's law and the emf of a cell (Plus Two; pre-physics P1).
- Separating variables and integrating a first-order equation (Beyond; pre-math M4). Learn the steps here before the derivation.
- Exponentials and logarithms (pre-math M3).

1. The circuit

Take an inductance L and a resistance R in series with a cell of steady emf E. A key K closes the circuit (Fig. 12.1). When the key is pressed, the current does not jump to its final value. It grows from zero to a maximum I_0. Let I be the current at time t.

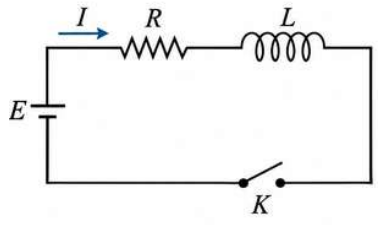


Fig. 12.1 (book): inductance L and resistance R in series with a cell E and key K. Schematic.

2. The loop equation

The inductor opposes the change of current with the back emf $\varepsilon = -L\,dI/dt$. Going round the loop, the cell emf is balanced by the drop across R and by the back emf of L:

$$E = RI + L\frac{dI}{dt} \qquad (12.1(1))$$

3. The steady current

When the current reaches its maximum I_0, it no longer changes, so $dI/dt = 0$ and the back emf is zero. Eq. (1) gives

$$E = RI_0 \qquad (12.1(2))$$

> **RECALL (PLUS TWO)** Ohm's law. With no change of current, the inductor is a plain wire, so the final current is $I_0 = E/R$. This is the same result the Plus Two circuit gives.

4. Separating the variables

Substitute $E = RI_0$ into Eq. (1). Then $RI_0 = RI + L\,dI/dt$, so

$$R(I_0 - I) = L\frac{dI}{dt} \quad \Rightarrow \quad \frac{dI}{I_0 - I} = \frac{R}{L}\,dt$$

The left side contains only I and the right side only t. The two can be integrated separately.

5. Integrating and fixing the constant

Integrating both sides,

$$-\log(I_0 - I) = \frac{R}{L}t + C \qquad (12.1(3))$$

where C is the constant of integration. At $t = 0$, $I = 0$, so $-\log I_0 = C$. Put this back into Eq. (3):

$$\log_e \frac{I_0 - I}{I_0} = -\frac{R}{L}t \quad \Rightarrow \quad 1 - \frac{I}{I_0} = e^{-(R/L)t}$$

6. The result

$$I = I_0\left(1 - e^{-(R/L)t}\right) \qquad (12.1(4))$$

Eq. (4) is the current in the L-R circuit at any instant. It starts at zero and tends to I_0 as t becomes large.

7. The time constant

The quantity L/R is the *time constant* τ of the circuit. Put $t = \tau = L/R$ in Eq. (4):

$$I = I_0\left(1 - e^{-1}\right) = 0.632\, I_0$$

So the time constant is the time the current takes to grow to 0.632 of its final value. A larger L/R means a slower growth. At $t = 2\tau$ the current is $0.865\, I_0$; at 3τ it is $0.950\, I_0$. Fig. 12.2 shows this for three values of L/R.

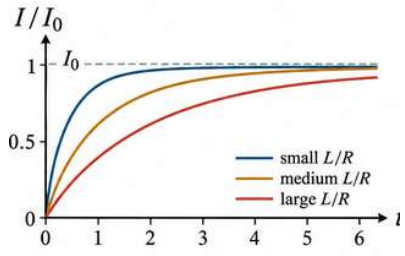


Fig. 12.2 (book): current I/I0 against time for low, medium and large L/R. Drawn from I = I0(1 − e^(−Rt/L)).

8. Worked example (printed p. 314)

An emf of 10 V is applied to a circuit with $R = 10\ \Omega$ and $L = 0.5$ H. The time to reach 63.2% of the final value is found from Eq. (4). Here $R/L = 20\ \text{s}^{-1}$, so $e^{-20t} = 0.368$ and $t = \frac{\ln 2.717}{20} = 0.05$ s. The time constant is $L/R = 0.05$ s, which is the same value. This is no accident: 63.2% is reached at $t = \tau$ by definition.

12.2 Decay of current in a circuit with inductance and resistance

Source: book-sources/m4-s12-2-decay.html, Murugeshan pp. 313–314 (printed), with Examples 1 and 2. Syllabus item 16. Equations (1) and (2) are given in full. Example 1 is a growth example; it is worked in §12.1 of these notes.

Before you start

- §12.1 of these notes: the loop equation and the time constant L/R.
- Exponentials and logarithms (pre-math M3).

1. What changes when the cell is removed

Once the current has reached I_0, the circuit is broken from the cell and closed on itself. The emf of the cell is now zero, so $E = 0$. The inductor still drives a current, because its back emf $-L\,dI/dt$ opposes the fall of current. The back emf keeps the current from dropping at once.

2. Loop equation with E = 0

$$0 = RI + L\frac{dI}{dt} \qquad (12.2(1))$$

Rearranging, $dI/I = -(R/L)\,dt$. The variables are separated again.

3. Integration

Integrate, with $\log_e I = -(R/L)t + C$. At $t = 0$, $I = I_0$, so $C = \log_e I_0$. Then

$$\log_e \frac{I}{I_0} = -\frac{R}{L}t \quad \Rightarrow \quad I = I_0 e^{-(R/L)t} \qquad (12.2(2))$$

4. Reading Eq. (2)

The current falls exponentially. The time constant $\tau = L/R$ is now the time for the current to fall to $1/e \approx 0.368$ of its initial value. The rate of decay is

$$\frac{dI}{dt} = -\frac{R}{L}I_0 e^{-(R/L)t} = -\frac{R}{L}I$$

The larger the ratio R/L, the more rapidly the current dies away (Fig. 12.4). The decay and growth curves are complementary (Fig. 12.5). The decay curve is the growth curve turned upside down.

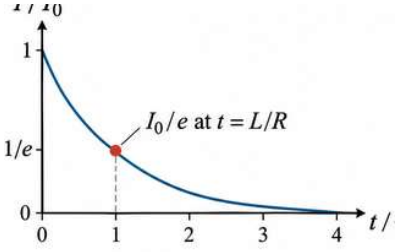


Fig. 12.3 (book): decay of current I = I0 e^(−Rt/L) against time. At t = L/R the current is I0/e.

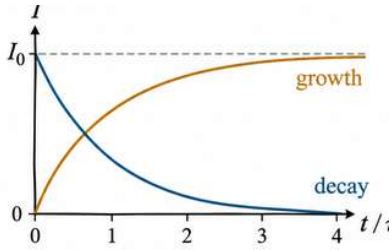


Fig. 12.5 (book): growth and decay curves shown together; they are complementary.

5. Worked example (printed p. 315)

An inductance of 500 mH and a resistance of 5 Ω are connected to a 10 V emf. The final current is $I_0 = E/R = 2$ A. When the cell is removed and the terminals are joined, $R/L = 10\ \text{s}^{-1}$ and $I = 2e^{-10t}$. After 0.05 s, $I = 1.213$ A. After 0.2 s, $I = 0.271$ A.